

Mini-Test
Modern Physics

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Text and solution for Monday's mini-test.

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WEEK II

- A gold cathode with work function $\phi_{\text{Au}} = 5.10 \text{ eV}$ is irradiated with UV light of wavelength $\lambda = 124 \text{ nm}$. What is the energy E_γ of the photons in electronvolts? (1pt) What is the stopping voltage V_0 for the emitted photoelectrons? (1pt)
Give the results in the boxes with three significant digits. Use the constant $hc = 1240 \text{ eV nm}$.

Solution

The energy of the photon is given by

$$E_\gamma = hf = h \frac{c}{\lambda} = 10.0 \text{ eV}.$$

The photoelectron maximum kinetic energy is given by

$$K_{\text{max}} = E_\gamma - \phi_{\text{Au}} = 4.90 \text{ eV}.$$

The stopping voltage is the applied (negative) potential difference that stops even the most energetic photoelectrons escaping the cathode from reaching the anode, thus

$$e|V_0| = K_{\text{max}}.$$

yielding

$$V_0 = 4.90 \text{ V},$$

where we used three significant digits and included the units for both answers.

- A potassium cathode with work function $\phi_{\text{K}} = 2.29 \text{ eV}$ is irradiated with UV light of wavelength $\lambda = 248 \text{ nm}$. What is the energy E_γ of the photons in electronvolts? (1pt) What is the stopping voltage V_0 for the emitted photoelectrons? (1pt)
Give the results in the boxes with three significant digits. Use the constant $hc = 1240 \text{ eV nm}$.

Solution

Analogously to the other version, the results are

$$E_\gamma = 5.0 \text{ eV},$$

and

$$K_{\text{max}} = 2.71 \text{ eV},$$

yielding

$$V_0 = 2.71 \text{ V},$$

where we used three significant digits and included the units for both answers.